Element C

Design Requirement #1 (HIGH Priority)

It must support Cooper and Abel's full weight without giving, meaning the braces of the legs don't wobble/move

A major issue of the old design is that the table doesn’t completely support the weight of students that wish to sit on the table. The design iterations represent a way to better support more weight.

Design Requirement #2 (HIGH Priority)

The design is replicable, can Mr.L read through the instructions/diagrams we provide him without any questions
Mr L. needs a replicable design. The in-depth iteration designs are a template for what we as the team must accomplish.

Design Requirement #3 (HIGH Priority)

When applied with a repeated amount of large force (someone hitting it with a hammer/mallet, jumping on it, something along those lines) x amount of times to the table, the legs don't move more than 1cm.

The diagrams of more stable tables directly connect to the necessity for the table to stay steady in times of stress.

Design Requirement #4 (HIGH Priority)

It must have enough clearance to fit a chair base fully under it (pushed in)

The folding and retractable leg design shows an option for our team to consider. The designs that distribute the load adequately often fail to consider this requirement.

Design Requirement #5 (HIGH Priority)

It must be easily fixable. If screws are loosened/leg falls off, it must take less than 1 minute to put back on.

Keeping the screws from coming out is a paramount issue in our design. If the design is screw free, it would be easily fixable, not needing Mr L. to waste time.

Design Requirement #6 (LOW Priority)

It looks nice and is aesthetic/matches with the rest of the room and doesn't look out of place

Considering the design of the skirt in the previous issues, making an aesthetic table can still be accomplished with distributing weight.

Design Requirement #7 (LOW Priority)

It should be able to roll on wheels, that can lock while no providing any interference with stability

The patent for the locking mechanism on the wheel can be used to show design requirement 7. The wheel can be iterated upon to better fit our needs, possibly to make it removable.